Fall 2013 / EC571

Instructor:Ajay Joshi

Mid-term Exam

Date: October 22, 2013

Name : _____

BU ID : _____

Question	Maximum points	Points scored
1	20	
2	20	
3	20	
4	20	
5	20	
6	20	
Bonus	5	
Total	100	

1. The duration of the exam is 2 hours.
2. The maximum grade is 100 points.
3. Please write your name on all pages.
4. Attempt any 5 of the 6 questions. Please indicate the question number that should not be graded.
5. The solutions should be written in the spaces provided on the test sheet.
6. If you need to use extra sheets of paper, write your name and BU ID on it, and clearly indicate the question number.
7. This is an open book and open notes exam.
8. Material sharing is not allowed.
9. Calculators are allowed.
10. Boston University Academic Code of Conduct applies.

Question 1

Draw schematics for the static complementary logic, pass transistor logic and pseudo-NMOS logic implementations of the following Boolean functions (assume you have both complemented and non-complemented inputs):

1. $f(A,B,C) = \overline{A \cdot B \cdot C}$. (7 points)

2. $f(J,K,L) = J \oplus K$. (8 points)

3. $f(E,F,G,H) = \overline{E \cdot F + G}$. (7 points)

Question 2

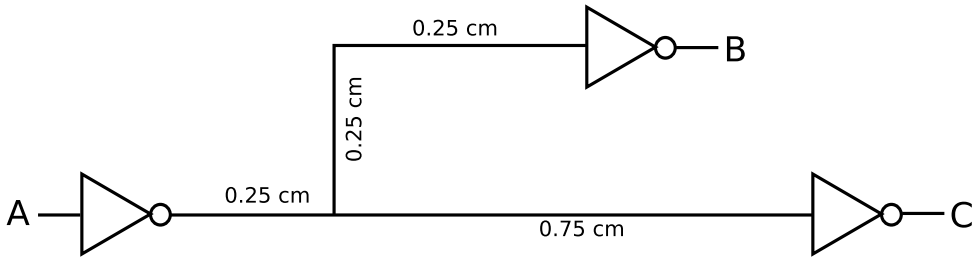
1. Draw the static complementary CMOS logic implementation for the following Boolean logic - $f(P,Q,R,S,T,U,V) = \overline{P+Q+R\cdot S+T\cdot(U+V)}$. (5 points)

2. Design the equivalent RC network for the circuit. You can assume the circuit drives a load capacitance of C_L. (5 points)

3. Determine the widths of the various transistors as a function of W so that worst case $\tau_{HL} = \tau_{LH}$. Use Elmore delay equations. You can assume $2R_{sq}(NMOS) = R_{sq}(PMOS)$. (5 points)

4. Calculate τ_{HL} and τ_{LH} (Elmore delay) using the following parameters – W = 1.5 micron, L = 0.25 micron $R_{sq}(NMOS)$ = 13 Kohm/sq, $R_{sq}(PMOS)$ = 26 Kohm/sq, C_g = 1 fF/micron, C_d = 1.5 fF/micron, C_L = 25 fF. (5 points)

Question 3

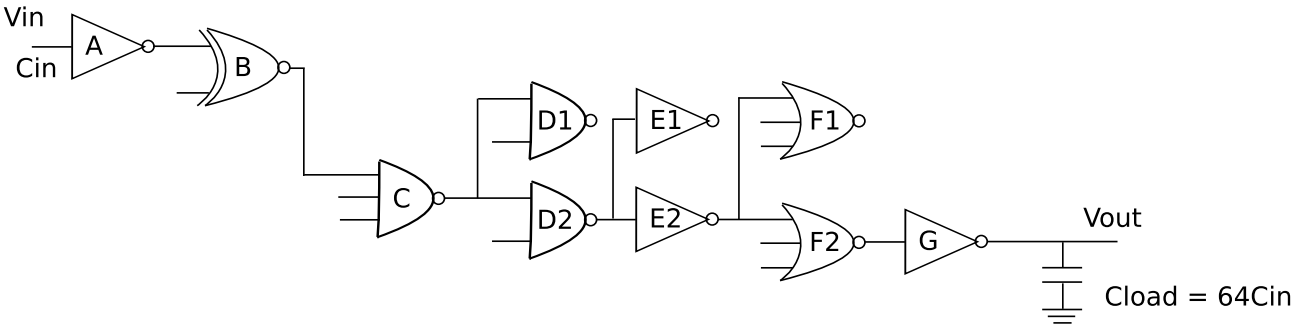


1. For the circuit shown in the above figure, draw the equivalent RC network that includes all the paths. Assume a lumped model for the different segments of the interconnect (7 points)

2. Write the Elmore delay equations for all three paths – A to B and A to C.(7 points)

3. Calculate the Elmore delay for all three paths – A to B and A to C. You can use the following technology parameters for calculating the delay - Vdd = 2.5 V, L = 0.25 micron Rsq(NMOS) = 13 Kohm/sq, Rsq(PMOS) = 26 Kohm/sq, Cg = 1.5 fF/micron, Cd = 2 fF/micron, wire capacitance = 0.2 fF/micron, wire resistance = 10 ohms/sq. The interconnect has a width of 1 micron. Assume that all four inverters use a minimum width of 300 nm for the NMOS. You will need to choose the width of the PMOS transistor such that the low-to-high delay of the inverters matches with the high-to-low delay. Assume a load of 10 fF at each output. (8 points)

Question 4



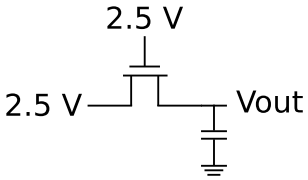
For the circuit shown in the above figure, using the logical effort technique, calculate the input width of transistors for each gate in the path such that total delay of the path is minimized. You can assume that gates D1 and D2 are identical, E1 and E2 are identical and gates F1 and F2 are identical. Assume Cin = 4 fF, Cg = 1 fF/u. (20 points)

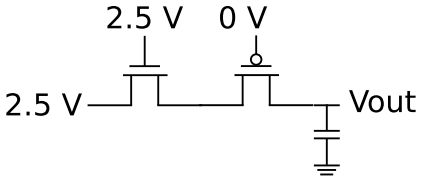
Question 5

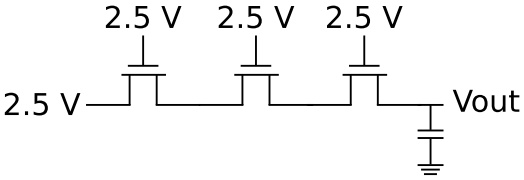
Design the static complementary CMOS implementation of a 2:1 multiplexer circuit for minimum delay (assuming a fanout i.e. stage effort of 4) and driving a load of 10 fF. As part of the design you need to determine the width of all transistors. You can use the following transistor parameters for the design - L = 0.25 micron $R_{sq}(NMOS)$ = 13 Kohm/sq, $R_{sq}(PMOS)$ = 26 Kohm/sq, C_g = 1 fF/micron, C_d = 1.5 fF/micron. You can assume that you have both complemented and non-complemented inputs (i.e. you have $A, \overline{A}, B, \overline{B}, Select, \overline{Select}$ and do not need additional inverters to generate these signals) and that input signals have very small rise and fall times. *Hint: Write the Boolean expression for the 2:1 multiplexer and use the Boolean expression to design the circuit. Use logical effort technique to determine the transistor sizes.* (20 points)

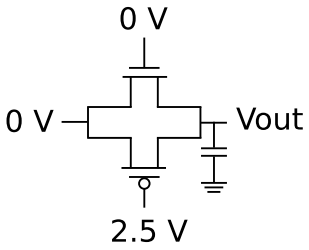
Question 6

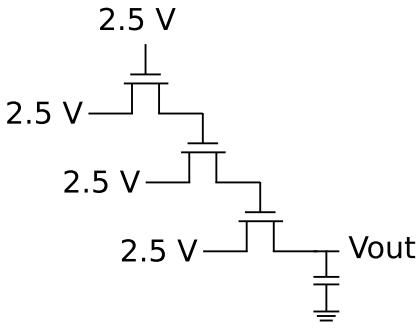
Find the final value of the voltage V_{out} for the following 5 configurations (4 points each). Assume V_{dd} = 2.5 V, $V_{th,N}$ = 0.5 V and $V_{th,P}$ = -0.5 V. Assume that the capacitor is initially discharged. Ignore subthreshold conduction and body effect.











Bonus question (5 points)
Prof. Joshi's 3-year old son has a Halloween party at his school. Please suggest five different costumes ideas for him.